Ultrasound – from sonar to scanning

Ultrasound is sound above the upper limit of hearing, about 20 kHz. With frequencies from a few hundred kHz to over 50 MHz the wavelength becomes so short that sound can be used like a torch: it is sent in a narrow beam into the body or a workpiece and reveals interfaces and flaws by their echoes. The page goes through the physics and has an interactive sound field from a transducer.

> DTH, course 1846 (1985), 6 points: Ultrasound. The course was closely related to Non-destructive testing and used the same concepts as audible acoustics – only with millimetre wavelengths and in solids and liquids.

From Curie to the scanner

The brothers Pierre and Jacques Curie discovered in 1880 that certain crystals such as quartz produce a voltage when compressed – *piezoelectricity*. During the First World War Paul Langevin used quartz crystals to build the first sonar for detecting submarines; the sinking of the Titanic in 1912 had prompted the idea of looking for icebergs by echo. Sergei Sokolov proposed in 1929 to find flaws in metal with ultrasound. After the Second World War Ian Donald in Glasgow transferred the technique from the shipyard flaw detector to obstetrics: his paper in *The Lancet* in 1958 began medical ultrasound scanning, today the most widely used imaging technique in the world.

Piezoelectric transducers

An ultrasonic transducer is a thin disc of piezoceramic, typically lead zirconate titanate (PZT), with electrodes on both sides. A voltage makes the disc change thickness; an echo pressing on the disc gives a voltage. The disc vibrates most strongly at its thickness resonance, where the thickness is half a wavelength in the ceramic: $f_0 = c_{\mathrm{PZT}} / 2d$. A 5 MHz transducer has a disc of about 0.4 mm. Behind the disc is a heavy damping backing, so that the pulse is short, and in front a matching layer a quarter wavelength thick with an impedance between that of the ceramic and the tissue – exactly like the anti-reflection coating on a camera lens.

Reflection at interfaces

At normal incidence on a plane interface between two media with acoustic impedances $Z_1 = \rho_1 c_1$ and $Z_2 = \rho_2 c_2$ the fractions of energy reflected and transmitted are

$$R_I = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2, \qquad T_I = \frac{4 Z_1 Z_2}{(Z_1 + Z_2)^2}$$

Interface	Z_1 / MRayl	Z_2 / MRayl	Reflected energy
Fat → muscle	1.38	1.70	1 %
Soft tissue → bone	1.63	7.8	43 %
Soft tissue → air	1.63	0.0004	99.9 %
Water → steel	1.48	45	88 %
Steel → air (a crack)	45	0.0004	≈ 100 %

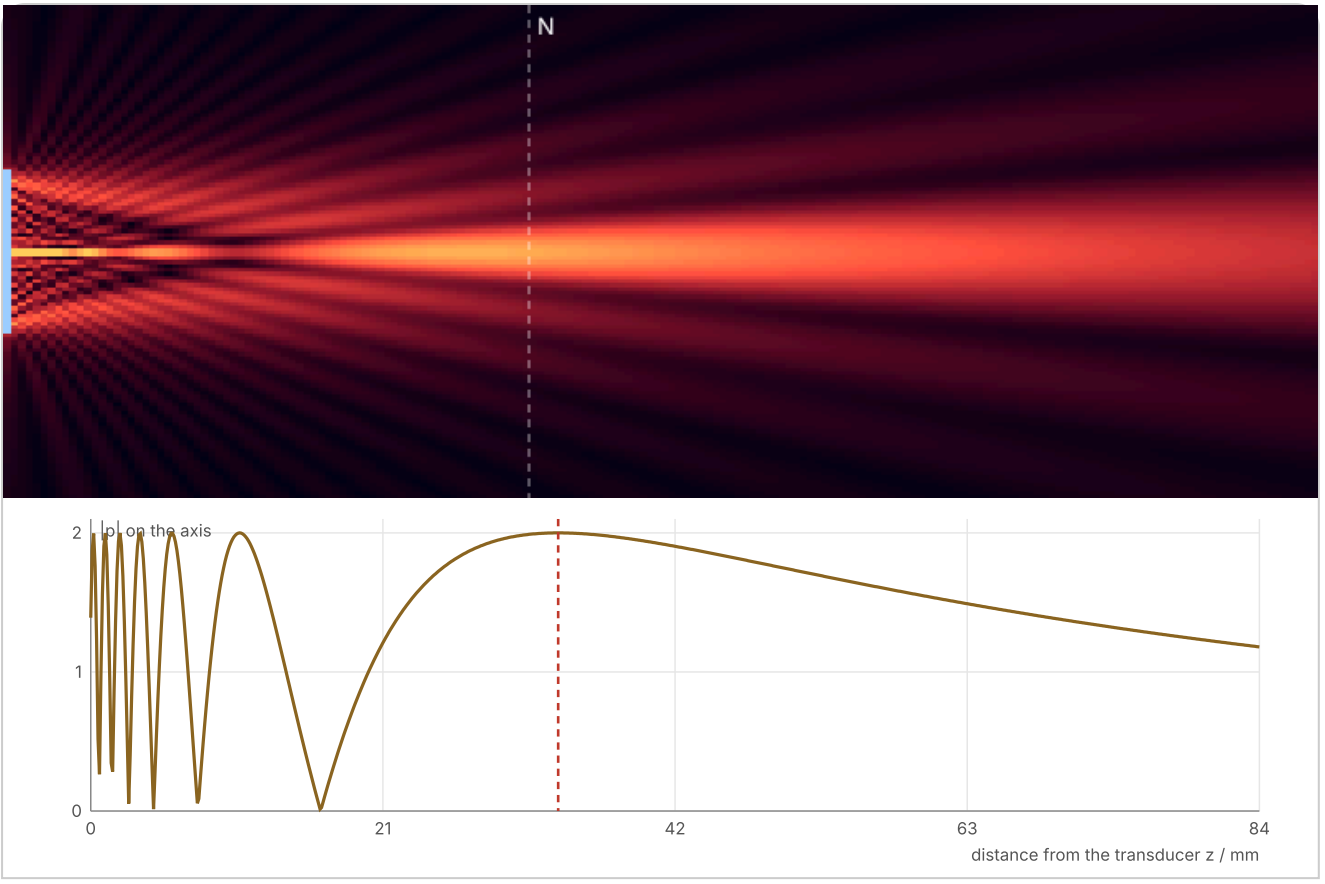
The small reflection between soft tissues is exactly what gives a picture of the organs – while letting most of the sound go on. Air reflects almost everything. That is why gel is spread on the skin, why one cannot see through the lungs, and why a crack in steel – a thin air gap – is so easy to find.

The sound field of a transducer

A plane, circular transducer with diameter D is a piston source, and every point on its face emits a wavelet (Huygens). Close to the transducer the waves from the centre and the edge interfere, and the sound pressure along the axis oscillates between zero and maximum. The last maximum lies at the near-field length

$$N = \frac{D^2 - \lambda^2}{4\lambda} \approx \frac{D^2}{4\lambda}, \qquad \sin\theta_0 = 1.22\,\frac{\lambda}{D}$$

Beyond N is the far field, where the beam spreads in a cone with half-angle θ_0 to the first zero – the same formula as for the diffraction of light through a circular aperture (Airy's disc). The beam is narrowest around N, and transducers are often focused with an acoustic lens or a curved face to make the beam even narrower at the depth of interest.



Attenuation

Ultrasound is attenuated by absorption (viscosity, heat conduction and relaxation) and by scattering from grains and inhomogeneities. In soft tissue the attenuation is roughly proportional to frequency, about 0.5 dB/(cm · MHz). A 3 MHz pulse that must travel to a depth of 15 cm and back is attenuated by $0.5 \cdot 3 \cdot 30 = 45$ dB. A higher frequency gives better resolution but less penetration – the fundamental trade-off in all ultrasound. That is why 2–5 MHz is used for the abdomen and foetuses, 7–15 MHz for muscles, tendons and the thyroid, and 20–50 MHz for skin and eyes. Scanners compensate for the attenuation with a gain that increases with time after the pulse (time-gain compensation).

Pulse–echo and resolution

The distance to a reflector is found from the echo's travel time: $d = ct/2$. Scanners assume $c = 1540$ m/s in tissue, so 13 µs corresponds to 1 cm. Two reflectors one behind the other can be distinguished if they are more than half a pulse length apart; with a pulse of two cycles at 5 MHz the axial resolution is about 0.3 mm. The lateral resolution is the beam width, typically 1–2 mm. A B-scan ("brightness") image is built up from many A-scan lines side by side; today with an array transducer, where the beam is steered and focused electronically by delaying the signals to the individual elements (phased array).

Doppler

Sound reflected from moving blood cells has its frequency shifted: $\Delta f = 2 f_0 v \cos\theta / c$. At 5 MHz and a blood velocity of 50 cm/s the Doppler shift is about 3 kHz – in the audible range, so the doctor can listen to it. Colour Doppler, which codes velocity in colours on top of the B-image, was introduced in the mid-1980s and is now standard in examinations of the heart and blood vessels.

High-power ultrasound

At high intensities ultrasound can produce *cavitation*: small bubbles that grow in the rarefaction phase and collapse violently, with local temperatures of thousands of degrees. This is used in ultrasonic cleaning baths, plastic welding, emulsification and the crushing of kidney stones. High-intensity focused ultrasound (HIFU) can heat a small region deep in the body and destroy a tumour without surgery.

Since 1989

- **3D and 4D:** Real-time volume scanning of foetuses came in the 1990s and is now common.
- **Pocket scanners:** Whole scanners have shrunk to a transducer connected to a mobile phone, used by general practitioners and paramedics.
- **Elastography** measures tissue stiffness by tracking shear waves – to assess cirrhosis of the liver and lumps in the breast.
- **Plane-wave and ultrafast imaging** with thousands of images per second; Jørgen Arendt Jensen at DTU developed the simulation program Field II, used in ultrasound research worldwide, and methods to measure blood velocity in all directions (vector flow).
- **Super-resolution:** By tracking individual microbubbles of contrast agent, blood vessels down to 10 µm can now be imaged – far below the wavelength.

Further reading

- Thomas L. Szabo: *Diagnostic Ultrasound Imaging: Inside Out* (2nd ed., Academic Press 2014)
- Josef and Herbert Krautkrämer: *Ultrasonic Testing of Materials* (4th ed., Springer 1990)
- Peter N. T. Wells: “Ultrasound imaging”, *Physics in Medicine and Biology* 51 (2006) R83–R98